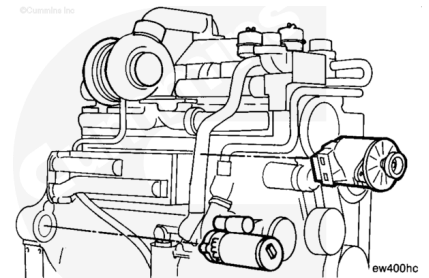


Trainings ▾

Install

Industrial and Power Generation

Install all accessories and brackets that were removed from the engine.

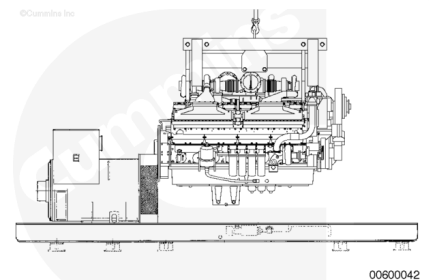


The engine lifting fixtures, Part Number 3163264, are designed to lift the engine **only**. Any additional components attached (including to the drive end, non-drive end and/or chassis) **must** be removed.

Use the following procedure to determine engine weights, Refer to Procedure 018-015 in Section V.
(/qs3/pubsys2/xml/en/procedures/56/56-018-015-tr.html)

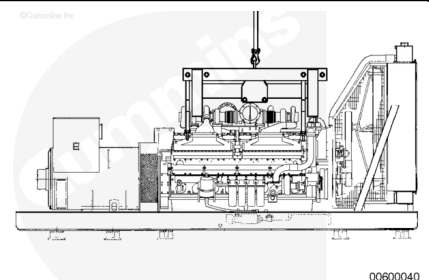
The engine lifting equipment is designed to lift the engine safely. Do **not** lift the engine and drivetrain as an assembly.

The lifting fixture, Part Number 3163264, is designed to lift a maximum of 11,340 kg [25,000 lb].



⚠ WARNING ⚠

Inspect the engine lifting brackets for damage or cracks. Do not attempt to lift the engine if any cracks or damage are visible. Using damaged lifting equipment can cause serious personal injury and/or property damage.



⚠ WARNING ⚠

The QSK45 and QSK60 series engines can weigh as much as 9980 kg [22,000 lb]. Use a properly rated hoist and engine lifting fixture, Part Number 3163264, to lift the engine.

Rigging and lifting must be done by trained, experienced personnel.

⚠ CAUTION ⚠

Make sure all lines, hoses, and tubes are in good condition, properly routed, and fastened to prevent damage.

Use a properly rated hoist and engine lifting fixture, Part Number 3163264, to install the engine.

The lifting force **must** be in a straight vertical line with the lifting brackets in both directions.

Align the engine in the chassis, and tighten the engine mounting capscrews. Refer to the OEM's torque specifications.

Connect all engine and chassis-mounted accessories that were removed.

With the engine installed check the crankshaft end clearance to make sure it is still within specification. Refer to Procedure 001-007 in Section 1.

(/qs3/pubsys2/xml/en/procedures/56/56-001-007-tr.html)

Remove an accessible hand hole cover. Refer to Procedure 001-035 in Section 1

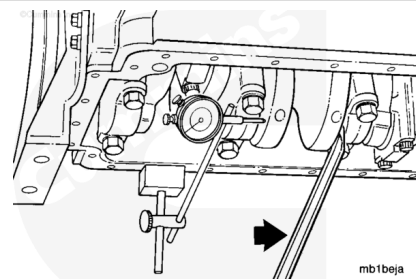
(/qs3/pubsys2/xml/en/procedures/56/56-001-035-tr.html)

The crankshaft can be moved forwards and backwards by using a pry bar on the crankshaft webs through the hand hole cover opening. The end clearance can be measured through the hand hole cover opening on a crankshaft counterweight or on the front damper.

Note : Do not use a pry bar on the damper to move the crankshaft. Doing so may cause damage to the damper.

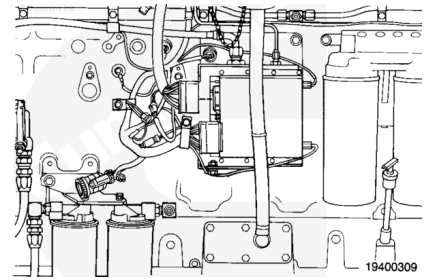
Note : The oil pan is shown removed from the engine for illustration purposes.

Once the end clearance is confirmed to be within specification replace the hand hole cover. Refer to Procedure 001-035 in Section 1 (/qs3/pubsys2/xml/en/procedures/56/56-001-035-tr.html). If the end clearance is out of specification, check for misalignment with the attached drive units.



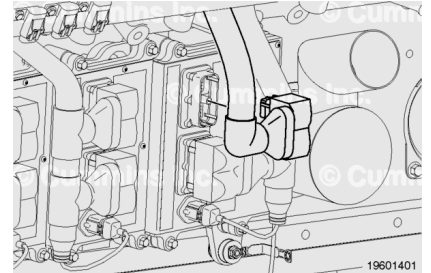
Connect the OEM wiring harness to the ECM.

- For QSK45 engines, refer to the Troubleshooting and Repair Manual, QSK Fuel System, QSK19, QSK23, QSK45, QSK60, and QSK78 Engines, Bulletin 3666113.



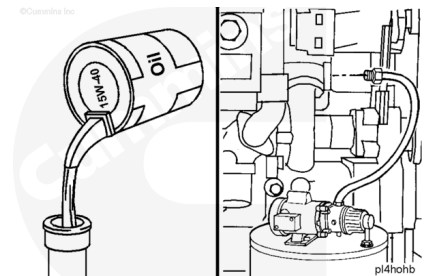
Electronically Actuated Injector

- For QSK60 engines, refer to the Troubleshooting and Repair Manual, QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System), Bulletin 4021533.



⚠ CAUTION ⚠

To prevent the possibility of engine damage the engine must be prelubricated by pumping pressurized oil through the system prior to starting the engine.



Fill the engine with clean 15W-40 lubricating oil. Refer to Procedure 007-037 in Section 7.

([/qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html](http://qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html))

Determine the actual capacity by oil pan part number. Refer to Procedure 018-017 in Section V.

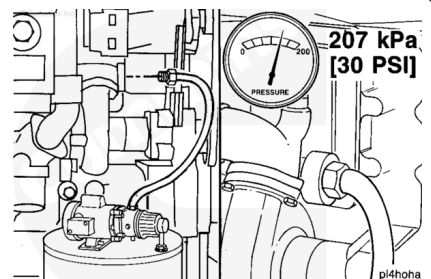
([/qs3/pubsys2/xml/en/procedures/56/56-018-017.html](http://qs3/pubsys2/xml/en/procedures/56/56-018-017.html))

Industrial engines are equipped with prelubrication devices from the factory. Engines other than industrial often do **not** have a prelubrication option. Follow the instructions below for the remote prelubrication process.

⚠ CAUTION ⚠

The lubricating oil system must be primed before operating the engine after rebuild to avoid internal component damage.

Use a pump capable of supplying 205-kPa [30-psi] continuous pressure. Connect the pump to the front of the oil filter head.

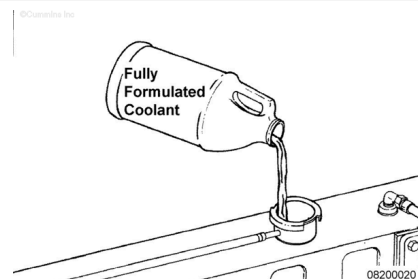


Use a supply of clean oil. Turn the pump to the ON position. Check the engine oil pressure gauge. When the gauge indicates oil pressure, begin monitoring the oil level in the oil pan.

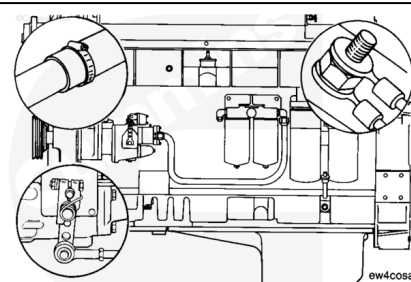
Fill the cooling system with fully formulated coolant. Information is found in the Operation and Maintenance Manual, QSK45 and QSK60 Engines, Bulletin 3666260. Refer to Procedure 018-018 in Section 8.

(/qs3/pubsys2/xml/en/procedures/56/56-018-018.html)

Refer to the OEM's specifications for radiator and system capacity.

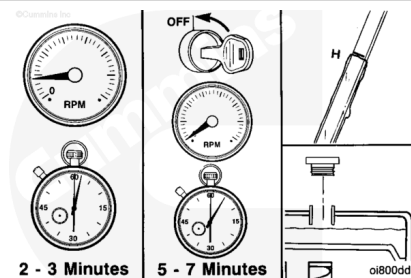


Perform a final inspection to make sure all hoses, wires, linkages, and components have been properly installed and tightened.



⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



Operate the engine at low idle for 2 to 3 minutes.

Stop the engine and wait 5 to 7 minutes for the oil to drain to the oil pan. Check the oil and coolant levels.

Finishing Steps

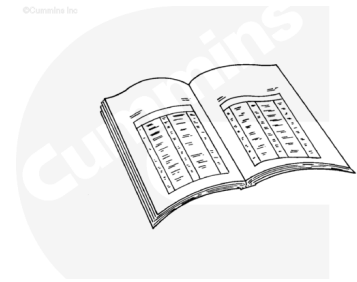
Industrial and Power Generation

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam

can cause personal injury.

- Perform a final inspection to make sure all hoses, wires, linkages, and components have been properly installed and tightened.
- Operate the engine at low idle for 2 to 3 minutes. Refer to Procedure 014-006 in Section 14.
(/qs3/pubsys2/xml/en/procedures/56/56-014-006-tr.html)
- Stop the engine and wait 5 to 7 minutes for the oil to drain to the oil pan. Check the oil and coolant levels.
- Add oil or coolant to obtain the correct level if necessary. Refer to Procedure 018-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-018-018.html)
Refer to Procedure 018-017 in Section V.
(/qs3/pubsys2/xml/en/procedures/56/56-018-017.html)
- Operate the engine for 8 to 10 minutes to check for proper operation, unusual noises, and coolant, fuel, or lubricating oil leaks. Repair all leaks and component problems. Refer to Procedure 014-006 in Section 14.
(/qs3/pubsys2/xml/en/procedures/56/56-014-006-tr.html)



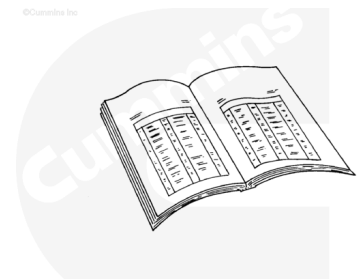
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QSK60 Marine Applications

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

- Install the turbocharger inlet connections. Refer to Procedure 010-130 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-130-tr.html)
- Install the turbocharger heat shields. Refer to Procedure 010-076 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-076-tr.html)
- Install the aftercooler heat shield. Refer to Procedure 010-129 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-129-tr.html)



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- Perform a final inspection to make sure all hoses, wires, linkages, and components have been properly installed and tightened.
- Operate the engine at low idle for 2 to 3 minutes. Refer to Procedure 014-006 in Section 14.
(/qs3/pubsys2/xml/en/procedures/56/56-014-006-tr.html)
- Stop the engine and wait 5 to 7 minutes for the oil to drain to the oil pan. Check the oil and coolant levels.
- Add oil or coolant to obtain the correct level if necessary. Refer to Procedure 018-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-018-018.html)
Refer to Procedure 018-017 in Section V.
(/qs3/pubsys2/xml/en/procedures/56/56-018-017.html)
- Operate the engine for 8 to 10 minutes to check for proper operation, unusual noises, and coolant, fuel, or lubricating oil leaks. Repair all leaks and component problems. Refer to Procedure 014-006 in Section 14.
(/qs3/pubsys2/xml/en/procedures/56/56-014-006-tr.html)